

# Full Solution Packet

## Alternating operator scaling has the spectral-gap rate

Run `fa_banach_001`, lane 1

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**Status:** candidate full solution, likely valid

**Target:** the open question after Corollary 1.6 of arXiv:1904.03213

### 1 Source and exact question

The source is Kwok–Lau–Ramachandran, *Spectral Analysis of Matrix Scaling and Operator Scaling* [1]. For a tuple  $\mathcal{A} = (A_1, \dots, A_k)$ ,  $A_i \in \mathbb{R}^{m \times n}$ , put

$$s(\mathcal{A}) = \sum_i \|A_i\|_F^2, \quad M_{\mathcal{A}} = \sum_i A_i \otimes A_i.$$

It is  $\epsilon$ -nearly doubly balanced when its two marginals lie between  $1 \pm \epsilon$  times their scalar targets, and it has a  $\lambda$ -spectral gap when

$$\sigma_2(M_{\mathcal{A}}) \leq (1 - \lambda) \frac{s(\mathcal{A})}{\sqrt{mn}}.$$

The source proves linear convergence of a gradient flow under  $\lambda^2 \geq C\epsilon \log m$  and then asks whether exact alternating left and right normalization has the same rate.

We prove that the gradient flow has linear convergence when the input satisfies the spectral gap condition.

**Theorem 1.5** (Linear Convergence). *Given an operator  $\mathcal{A} = (A_1, \dots, A_k)$  where each  $A_i \in \mathbb{R}^{m \times n}$  with  $m \leq n$ , if  $\mathcal{A}$  is  $\epsilon$ -nearly doubly balanced and  $\mathcal{A}$  satisfies the  $\lambda$ -spectral gap condition in Definition 1.4 with  $\lambda^2 \geq C\epsilon \log m$  for a sufficiently large constant  $C$ , then in the gradient flow,*

$$\Delta^{(t)} \leq \Delta^{(0)} e^{-\lambda s^{(0)} t} \quad \text{for any } t \geq 0.$$

*In particular, the gradient flow converges to a  $\eta$ -nearly doubly balanced scaling in time  $t = O\left(\frac{1}{\lambda} \log\left(\frac{m}{\eta}\right)\right)$ , and such a scaling always exists under our assumptions.*

By discretizing the gradient flow with step size  $\Theta((m+n)^{-2})$ , it follows that a natural gradient descent algorithm returns an  $\eta$ -nearly doubly stochastic scaling in polynomial time in the input size and logarithmic in  $1/\eta$ , when the input satisfies the spectral gap condition.

**Corollary 1.6** (Gradient Descent). *Under the assumptions in Theorem 1.5, there is a gradient descent algorithm to return an  $\eta$ -nearly doubly balanced scaling in  $O\left(\frac{(n+m)^2}{\lambda} \log\left(\frac{m+n}{\eta}\right)\right)$  iterations.*

It is an interesting open question whether the alternating algorithm also has the same convergence rate as the gradient flow under the same assumptions. We believe that the answer is positive but we could not prove it yet.

The question is transcribed as follows: “It is an interesting open question whether the alternating algorithm also has the same convergence rate as the gradient flow under the same assumptions.”

## 2 Result

The alternating algorithm replaces  $A_i$  successively by

$$\left(\sum_j A_j A_j^*\right)^{-1/2} A_i \quad \text{and} \quad A_i \left(\sum_j A_j^* A_j\right)^{-1/2},$$

with the harmless scalar factors that keep the size normalized. A full cycle means one left and one right normalization.

**Theorem 1** (Alternating spectral-gap convergence). *There are universal constants  $C, c > 0$  with the following property. Let  $m \leq n$ ,  $0 < \lambda \leq 1$ , and let  $\mathcal{A}$  be  $\epsilon$ -nearly doubly balanced with a  $\lambda$ -spectral gap. If*

$$\lambda^2 \geq C\epsilon \log(2m),$$

*then exact alternating normalization is well-defined and, after  $t$  full cycles, has a right-balanced iterate whose other normalized marginal error  $H_t$  satisfies*

$$\frac{\|H_t\|_F}{\sqrt{m}} \leq C\epsilon e^{-c\lambda t}.$$

*Consequently it returns an  $\eta$ -nearly doubly balanced scaling in*

$$O\left(\frac{1}{\lambda} \log \frac{m+n}{\eta}\right)$$

*alternating iterations. Thus the answer to the source question is positive.*

## 3 Proof intuition

Stop after every right normalization. Then the right marginal is exactly the identity and the left marginal is  $P = I + H$ . Normalize the completely positive map to an operator  $T$  between Hilbert–Schmidt matrix spaces. On the scalar identity directions,  $T$  is approximately the identity. The source’s spectral gap therefore forces the compression of  $T$  between the two traceless subspaces to have norm at most  $1 - c\lambda$ .

Left normalization changes the new right error, to first order, from  $H$  to  $-T^*H$ ; right normalization changes the next left error, to first order, to  $TT^*H$ . Hence one full cycle contracts normalized Frobenius error by  $1 - c\lambda$ . The inverse-square-root remainders cost only  $O(\epsilon)$ .

The remaining issue is global rather than local: could the marginal operator norm or the spectral gap deteriorate before contraction becomes effective? For the first  $O(\lambda^{-1} \log m)$  cycles, the operator error grows by at most  $1 + O(\epsilon)$  per cycle. The assumed inequality  $\epsilon \log m \ll \lambda^2$  makes the accumulated scaling  $O(\lambda)$ , so at least half the gap survives. By then Frobenius contraction controls the operator norm directly, and the argument continues to arbitrary accuracy.

## 4 One-cycle estimates

All matrix spaces below carry the Hilbert–Schmidt inner product. Write

$$e_m = I_m/\sqrt{m}, \quad e_n = I_n/\sqrt{n},$$

and let  $\Pi_m, \Pi_n$  denote orthogonal projection onto the corresponding traceless subspaces.

Globally rescale so that  $s(\mathcal{A}) = n$ , and suppose first that the right marginal is exactly balanced:

$$\Phi_{\mathcal{A}}^*(I_m) = I_n, \quad P := \frac{m}{n} \Phi_{\mathcal{A}}(I_n) = I_m + H.$$

Then  $H$  is traceless. Define

$$T = \sqrt{\frac{m}{n}} \Phi_{\mathcal{A}}.$$

Thus

$$T^*e_m = e_n, \quad Te_n = e_m + \frac{H}{\sqrt{m}}.$$

**Lemma 2** (Traceless spectral contraction). *Assume  $\|H\|_{\text{op}} \leq \delta$ ,  $\sigma_2(T) \leq 1 - \gamma$ , and  $\delta \leq c_0\gamma$  for a sufficiently small universal  $c_0$ . Then*

$$\|\Pi_m T \Pi_n\|_{\text{HS} \rightarrow \text{HS}}, \quad \|\Pi_n T^* \Pi_m\|_{\text{HS} \rightarrow \text{HS}} \leq 1 - c_1\gamma$$

for a universal  $c_1 > 0$ .

*Proof.* The largest-singular-value estimate from the source [1, Lemma 3.6] gives  $\sigma_1(T) \leq 1 + \delta$ . Moreover  $\langle e_m, Te_n \rangle = 1$ . Apply the elementary spectral lemma [1, Lemma 3.7]: if unit vectors  $p, q$  satisfy  $p^*Tq = 1$ ,  $\sigma_1(T)^2 \leq 1 + d_1$ , and  $\sigma_2(T)^2 \leq 1 - d_2$ , then

$$|x^*Ty| \leq 1 + d_1 - d_2 \quad (x \perp p, y \perp q).$$

Here  $d_1 \leq 3\delta$  and  $d_2 = 2\gamma - \gamma^2$ . Reducing  $c_0$  if necessary makes the last bound at most  $1 - c_1\gamma$ . Taking the supremum over traceless unit vectors proves both displayed estimates.  $\square$

**Lemma 3** (One full normalization cycle). *In the setting of Lemma 2, put*

$$\rho = \frac{\|H\|_F}{\sqrt{m}}.$$

*After exact left normalization followed by exact right normalization, let  $I_m + H_+$  be the new normalized left marginal, and put  $\delta_+ = \|H_+\|_{\text{op}}$  and  $\rho_+ = \|H_+\|_F/\sqrt{m}$ . If  $\delta \leq c_2\gamma$ , then*

$$\rho_+ \leq (1 - c_3\gamma)\rho, \quad \delta_+ \leq (1 + C_2\delta)\delta,$$

for universal constants  $c_2, c_3, C_2 > 0$ .

*Proof.* Left normalization uses  $L = P^{-1/2}$ . Its right marginal is  $I_n + K$ , where

$$K = \Phi_{\mathcal{A}}^*(P^{-1}) - I_n.$$

It is traceless: using  $\Phi_{\mathcal{A}}(I_n) = (n/m)P$ ,

$$\text{tr} \Phi_{\mathcal{A}}^*(P^{-1}) = \text{tr}(P^{-1} \Phi_{\mathcal{A}}(I_n)) = n.$$

The expansion

$$P^{-1} - I_m = -H + H^2 P^{-1}$$

and Lemma 2 give

$$\frac{\|K\|_F}{\sqrt{n}} \leq (1 - c_1\gamma + C_3\delta)\rho. \quad (1)$$

Indeed, the traceless part of  $T^*H$  contracts by Lemma 2; its scalar part has norm  $\rho\|H\|_F$ , because  $\langle e_n, T^*H \rangle = \langle Te_n, H \rangle = \|H\|_F^2/\sqrt{m}$ . This and the term  $H^2 P^{-1}$  are both absorbed by  $C_3\delta\rho$ .

Since  $\Phi_{\mathcal{A}}^*$  is positive and unital, it is an operator-norm contraction. Consequently

$$\|K\|_{\text{op}} \leq \|P^{-1} - I_m\|_{\text{op}} \leq \frac{\delta}{1 - \delta}. \quad (2)$$

Now right normalize by  $R = (I_n + K)^{-1/2}$ . The next left error is exactly

$$H_+ = P^{-1/2} \sqrt{\frac{m}{n}} T((I_n + K)^{-1} - I_n) P^{-1/2}.$$

Use  $(I + K)^{-1} - I = -K + K^2(I + K)^{-1}$ . Because  $K$  is traceless, Lemma 2 contracts the linear term. Equations (1)–(2), together with  $\|P^{-1/2}\|_{\text{op}}^2 \leq (1 - \delta)^{-1}$ , therefore yield

$$\rho_+ \leq (1 + C_4\delta)(1 - c_1\gamma + C_4\delta)^2\rho \leq (1 - c_3\gamma)\rho.$$

For the operator norm, positivity gives

$$\left\| \sqrt{\frac{m}{n}} T(X) \right\|_{\text{op}} \leq \|P\|_{\text{op}} \|X\|_{\text{op}}.$$

Using this with the exact formula for  $H_+$  and (2) gives

$$\delta_+ \leq \frac{1 + \delta}{1 - \delta} \frac{\delta}{1 - 2\delta} \leq (1 + C_2\delta)\delta,$$

as claimed.  $\square$

## 5 Gap persistence and proof of Theorem 1

First rescale to size  $n$  and perform one right normalization. Standard inverse-square-root estimates applied to the initial two marginals give a right-balanced tuple with

$$\delta_0, \rho_0 \leq C_5\epsilon, \quad \sigma_2(T_0) \leq 1 - \frac{3}{4}\lambda. \quad (3)$$

The last inequality follows because right normalization multiplies  $M_{\mathcal{A}}$  on the right by  $R \otimes R$  with  $\|R - I\|_{\text{op}} = O(\epsilon)$ . These estimates also show directly that every normalization used below is invertible.

Let  $\delta_j, \rho_j$  be the errors after the  $j$ th right normalization. As long as  $\delta_j \leq 2C_5\epsilon$  and the current gap is at least  $\lambda/2$ , Lemma 3 gives

$$\rho_j \leq C_5\epsilon e^{-c_3\lambda j}, \quad \delta_{j+1} \leq (1 + C_2\delta_j)\delta_j. \quad (4)$$

We record how the gap changes. A full cycle replaces the natural matrix by

$$M \mapsto (L \otimes L)M(R \otimes R),$$

where  $\|L\|_{\text{op}}^2, \|R\|_{\text{op}}^2 \leq 1 + C_6\delta_j$ . Since the size returns to  $n$  after every full cycle,

$$\sigma_2(T_j) \leq \sigma_2(T_0) \exp\left(C_7 \sum_{i < j} \delta_i\right). \quad (5)$$

Choose  $j_0 = C_8\lambda^{-1} \log(2m)$ , with  $C_8$  large enough that (4) implies  $\rho_{j_0} \leq C_5\epsilon/\sqrt{m}$ . On  $0 \leq j \leq j_0$ , the second inequality in (4) and a bootstrap give

$$\delta_j \leq 2C_5\epsilon, \quad \sum_{j < j_0} \delta_j \leq C_9 \frac{\epsilon \log(2m)}{\lambda}.$$

The hypothesis  $\lambda^2 \geq C\epsilon \log(2m)$ , with  $C$  sufficiently large, makes the last quantity at most  $c\lambda$ . Equation (5) then preserves a gap of at least  $\lambda/2$ , closing the bootstrap.

After  $j_0$ , operator norm is controlled directly by Frobenius norm:

$$\delta_j \leq \|H_j\|_F = \sqrt{m} \rho_j.$$

The remaining sum is geometric,

$$\sum_{j \geq j_0} \delta_j \leq \frac{\sqrt{m} \rho_{j_0}}{c_3\lambda} \leq C_{10} \frac{\epsilon}{\lambda} \log \lambda.$$

Thus the gap never falls below  $\lambda/2$ , and Lemma 3 proves (4) for all  $j$ .

At a right-balanced iterate, the other marginal has operator error at most  $\sqrt{m}\rho_j$ . Therefore  $\sqrt{m}\rho_j \leq \eta$  after

$$j = O\left(\frac{1}{\lambda} \log \frac{C\epsilon\sqrt{m}}{\eta}\right) = O\left(\frac{1}{\lambda} \log \frac{m+n}{\eta}\right)$$

cycles. One cycle has two alternating normalizations, so the same asymptotic bound holds for alternating iterations. This proves Theorem 1.

## 6 Verification report

- **Exact target.** The conclusion is the rate asked for after the source's Corollary 1.6, under the same near-balance and spectral-gap scale.
- **Dimension factors.** The normalizations  $T = \sqrt{m/n} \Phi_{\mathcal{A}}$ ,  $P = (m/n) \Phi_{\mathcal{A}}(I_n)$ , and  $e_d = I_d/\sqrt{d}$  make the scalar singular directions unit length; all  $\sqrt{m}, \sqrt{n}$  factors in the two half-steps cancel.
- **Nonlinear terms.** Both inverse expansions are exact identities, not first-order approximations. Their remainders are controlled by the operator error  $\delta$ .
- **Global bootstrap.** The only logarithmic dimension loss occurs before Frobenius error controls operator error. This is why the source's  $\epsilon \log m \ll \lambda^2$  hypothesis is sufficient.
- **Numerical sanity check.** The included script constructs a perturbed random balanced quantum expander. In one fixed run the marginal error decreased from  $6.55 \cdot 10^{-2}$  to  $1.25 \cdot 10^{-15}$  in 23 cycles; the spectral gap stayed above 0.4898, and the median cycle ratio was 0.202. The computation is not used in the proof.

## 7 Novelty and human review recommendation

The run’s registry, solution, attempt, and proof-gap indexes were searched for arXiv:1904.03213, operator scaling, Sinkhorn, alternating normalization, spectral gap, and quantum expander; no prior packet was found. A bounded external search used the exact question sentence and the phrases “operator Sinkhorn spectral gap convergence” and “quantum expander alternating normalization.” It found the source and the 2024 preprint of Soma–Uschmajew [2], which proves local asymptotic rates near a fixed point and studies overrelaxation, but does not state the global theorem above from a nearly balanced spectral-gap input. Novelty confidence is therefore moderate, not absolute.

Send for specialist review. The highest-value checks are: (i) the use of the source’s spectral lemma as a two-sided traceless compression bound, (ii) the scalar component of  $T^*H$  in (1), and (iii) the cumulative singular-gap estimate (5) for noncommuting successive scalings.

## References

- [1] T. C. Kwok, L. C. Lau, and A. Ramachandran, *Spectral Analysis of Matrix Scaling and Operator Scaling*, SIAM J. Comput. 50 (2021), no. 3, 1039–1102; arXiv:1904.03213.
- [2] T. Soma and A. Uschmajew, *Accelerating Operator Sinkhorn Iteration with Overrelaxation*, arXiv:2410.14104; Math. Program. (2026).